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% Exercícios Computacionais - Semana 1
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```
%% Questão 1 - Sinais Senoidais
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```
% Simulação dos sinais  $x(t) = A \cdot \sin(\omega t + \phi)$ , segue abaixo as alternativas:
```

```
% (a)  $A=5$ ,  $\omega=2\pi$ ,  $\phi=0$ :
```

```
t = 0:0.001:1; % t=1s
```

```
A = 5; w = 2*pi; phi = 0;
```

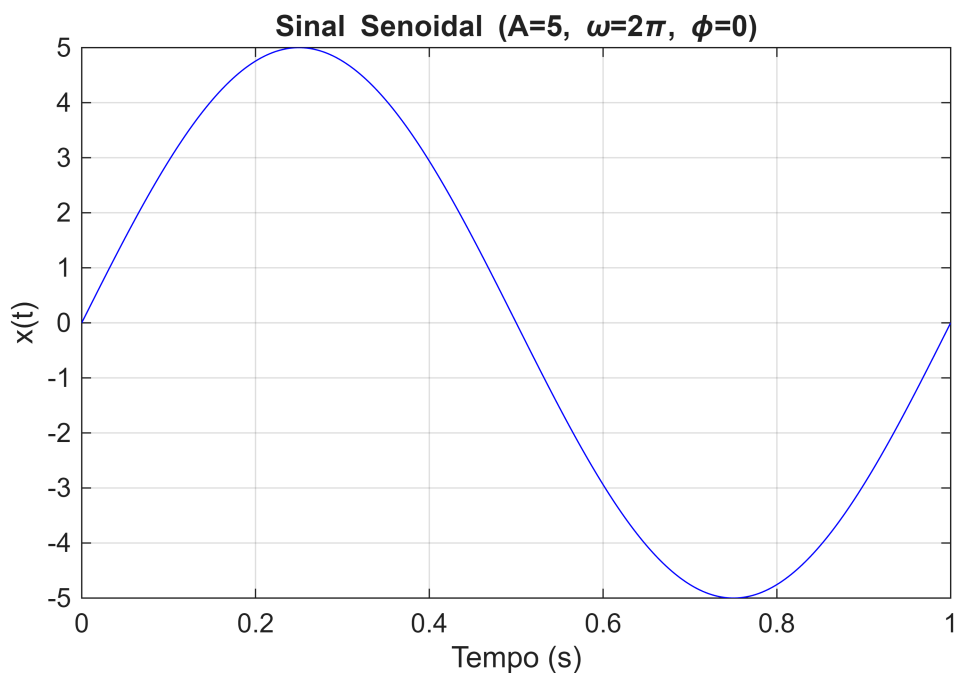
```
x1 = A * sin(w*t + phi);
```

```
figure;
```

```
plot(t, x1, 'b'); grid on;
```

```
title('Sinal Senoidal ( $A=5$ ,  $\omega=2\pi$ ,  $\phi=0$ )');
```

```
xlabel('Tempo (s)'); ylabel('x(t)');
```



```
% (b)  $A=10$ ,  $\omega=20\pi$ ,  $\phi=\pi/2$ 
```

```
t = 0:0.001:1; % mesmo intervalo de t
```

```
A = 10; w = 20*pi; phi = pi/2;
```

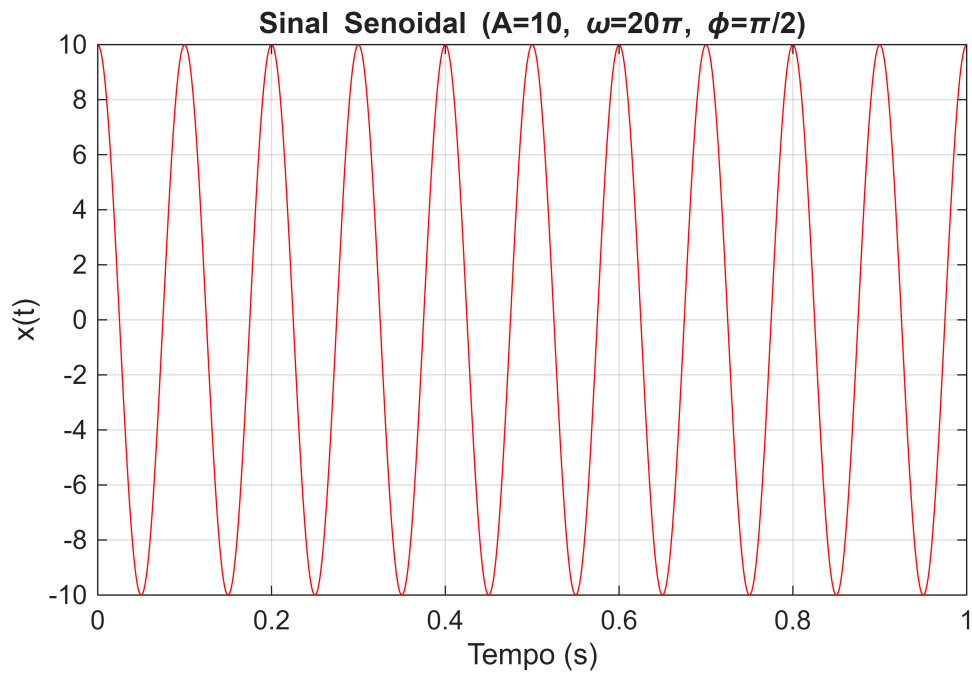
```
x2 = A * sin(w*t + phi);
```

```
figure;
```

```
plot(t, x2, 'r'); grid on;
```

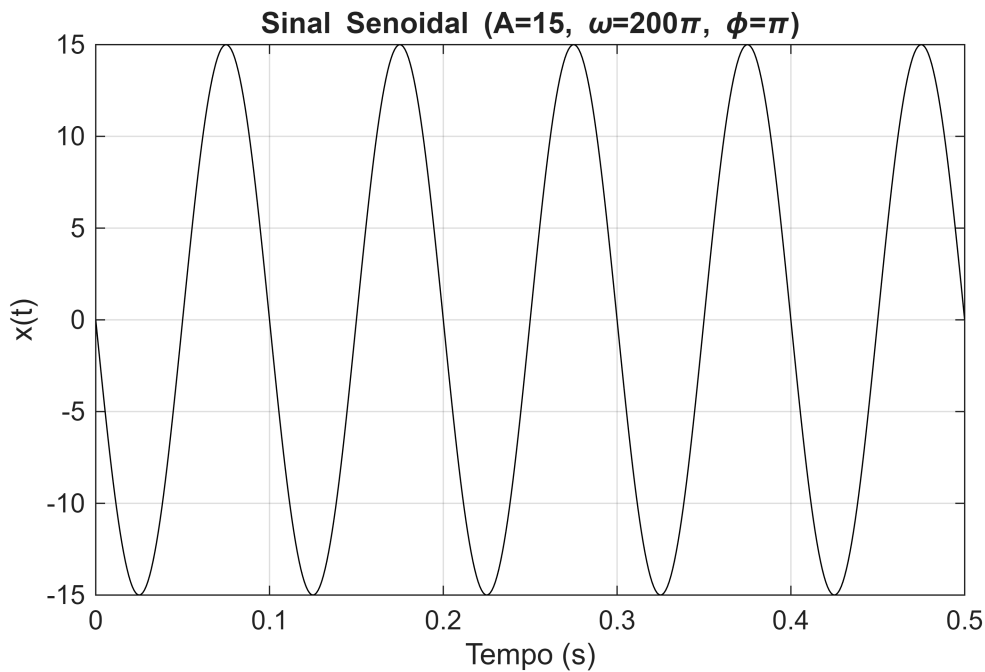
```
title('Sinal Senoidal ( $A=10$ ,  $\omega=20\pi$ ,  $\phi=\pi/2$ )');
```

```
xlabel('Tempo (s)'); ylabel('x(t)');
```



```
% (c) A=15, w=200\pi, \phi=\pi
% Frequência muito alta → passo menor e intervalo reduzido
T = (2*pi)/w;           % período do sinal
t = 0:T/100:5*T;       % 100 pontos por período, mostrando 5 períodos
A = 15; phi = pi;
x3 = A * sin(w*t + phi);

figure;
plot(t, x3, 'k'); grid on;
title('Sinal Senoidal (A=15, \omega=200\pi, \phi=\pi)');
xlabel('Tempo (s)'); ylabel('x(t)');
```



```
%% Questão 2 - Sinais Exponenciais
```

```
% Intervalo de tempo (0 a 10 segundos, passo 0.01), t=10s, ou seja, t∈[0,10]s
```

```
t = 0:0.01:10;
```

```
% (a)  $x(t) = e^{5t}$ 
```

```
x4 = exp(5*t);
```

```
% (b)  $x(t) = e^{-5t}$ 
```

```
x5 = exp(-5*t);
```

```
% (c)  $x(t) = e^{-t} \cdot \cos(10\pi t)$ 
```

```
x6 = exp(-t) .* cos(10*pi*t);
```

```
% Plotando em subplots
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot(t, x4, 'b'); grid on;
```

```
title('Sinal  $x(t) = e^{5t}$ ');
```

```
xlabel('Tempo (s)'); ylabel('x(t)');
```

```
subplot(3,1,2);
```

```
plot(t, x5, 'r'); grid on;
```

```
title('Sinal  $x(t) = e^{-5t}$ ');
```

```
xlabel('Tempo (s)'); ylabel('x(t)');
```

```
subplot(3,1,3);
```

```
plot(t, x6, 'k'); grid on;
```

```
title('Sinal  $x(t) = e^{-t}\cos(10\pi t)$ ');
```

```
xlabel('Tempo (s)'); ylabel('x(t)');
```

